

Segundo Parcial -

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④ $\rightarrow \phi(r) = -\frac{GM}{(r+a)}$

El perfil de Himmigat se usa para perfiles con simetría esférica

A partir de la ecuación de Poisson tenemos que:

$\nabla^2 \phi(r) = 4\pi G \rho(r)$, dada la simetría esférica del potencial $\nabla^2 \equiv \nabla^2(r) = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right)$

$\Rightarrow \frac{d}{dr} \phi(r) = \frac{GM}{(r+a)^2} \quad \frac{d}{dr} \left(\frac{GM r^2}{(r+a)^2} \right)$

$\Rightarrow GM \left(\frac{2r}{(r+a)^2} - \frac{2r^2}{(r+a)^3} \right)$

$\frac{2r GM}{r^2} \left(\frac{(r+a) - r}{(r+a)^3} \right) = \frac{2GMa}{r} \cdot \frac{1}{(r+a)^3}$

$\Rightarrow 4\pi \rho(r) = \frac{2GMa}{r} \frac{1}{(r+a)^3}$

$\rho(r) = \frac{M}{2\pi} \frac{a}{r} \frac{1}{(r+a)^3}$

Como la galaxia tiene una simetría esférica

$\rho(r) = \frac{dM(r)}{dV} \Rightarrow dM(r) = \rho(r) dV$

$M(r) = 4\pi \int_0^r dr r^2 \rho(r) = 4\pi \int_0^r dr r^2 \frac{M}{2\pi} \frac{a}{r} \frac{1}{(r+a)^3}$

$M(r) = 2Ma \int_0^r \frac{dr}{(r+a)^3} \quad u=r+a \Rightarrow r=u-a \quad du=dr$

$M(r) = 2Ma \left(\frac{r^2}{2a(a+r)^2} \right)$

$M(r) = 2Ma \left(\frac{r^2}{(a+r)^2} \right)$

La velocidad circular es:

$v_c^2 = r \frac{d\phi(r)}{dr} = r \frac{GM}{(r+a)} \Rightarrow v_c = \sqrt{\frac{r GM}{(r+a)}}$

La velocidad de escape

$v_{esc}(r) = \sqrt{2|\phi(r)|} = \sqrt{\frac{2GM}{(r+a)}}$

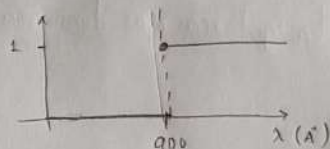
$\lambda = v_o \cdot c$

$\lambda \cdot v_o = c$

m/s

③

$F(\lambda) = \begin{cases} 0 & \text{si } \lambda < 900 \\ 1 & \text{si } \lambda \geq 900 \end{cases}$



Galaxia A: $dl = z \cdot c = 0.09174c$ $\lambda_c(A) = 824.37 \text{ \AA}$

Galaxia B: $dl = z \cdot c = 0.018c$ $\lambda_c(B) = 884.086 \text{ \AA}$

La longitud de onda deseada es $\lambda_o = 900 \text{ \AA}$

$v_o = v_c(1+z) \Rightarrow \frac{1}{\lambda_c} = \frac{1}{\lambda_o}(1+z)$

$v_o = c/\lambda_o$

$v_o = \frac{3 \times 10^8 \text{ m/s}}{900 \times 10^{-10} \text{ m}}$

$v_c = v_o(1+z)$

$\frac{\lambda_o}{1+z} = \lambda_c$

$v_o = 3.33 \times 10^{15} \text{ Hz}$

• Para (A) $v_c = v_o(1+0.09174) = 1.09174v_o$

• Para (B) $v_c = v_o(1+0.018) = 1.018v_o$

$F(v_o) = 1$

$K(z) = -2.5 \log(1+z) - 2.5 \log \left(\frac{L(v_o)}{L(v_c)} \right)$

$F(v_o) = \frac{L(v_o)}{4\pi d^2}$

$\Rightarrow L(v_o) = 4\pi d^2 F(v_o)$

$F(v_c) = \frac{L(v_c)}{4\pi d_c^2}$

$L(v_c) = \frac{4\pi d_c^2 F(v_o)}{(1+z)}$

$\frac{L(v_c)}{L(v_o)} = \left(\frac{d_c}{d} \right)^2 \frac{1}{(1+z)}$

$K(z) = -2.5 \log(1+z) - 2.5 \log \left(\left(\frac{d_c}{d} \right)^2 \frac{1}{(1+z)} \right)$

$K(z) = -5 \log \left(\frac{d_c}{d} \right) = -5 \log \left(\frac{d}{d} \cdot \frac{1}{c} \right) = 5 \log(c)$

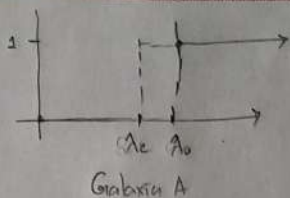
$m_B = M_B + 5 \log \left(\frac{d_c}{10 \text{ pc}} \right) - 5 \log \left(\frac{d}{d} \right)$

$M_B = m_B - 5 \log \left(\frac{d}{10 \text{ pc}} \right) - 5 \log \left(\frac{d}{d} \right)$

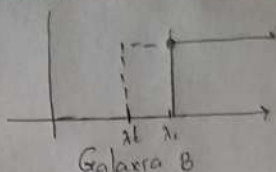
Galaxia A: $M_B = 17.18 - 5 \log \left(\frac{0.09174c}{10 \text{ pc}} \right) + 5 \log(c)$

Galaxia B: $M_B = 15.87 - 5 \log \left(\frac{0.018c}{10 \text{ pc}} \right) + 5 \log(c)$

⑤



Galaxia A



Galaxia B

$$M_B(A) = 69.783$$

$$M_B(B) = 76.974$$

La galaxia mas luminosa es la galaxia A

1) $R = R_g + x$ $H_R = \frac{1}{2} (\dot{R}^2 + k^2 (R_g - R)^2)$
a) $\dot{R} = \dot{x}$
 $\ddot{R} = \ddot{x}$ $H_z = \frac{1}{2} (\dot{z}^2 + v^2 z^2)$

$$P_R = m\dot{R} \quad P_z = m\dot{z}$$

$$H = \frac{1}{2m^2} (P_R^2 + P_z^2) + \frac{1}{2} (v^2 z^2 + k^2 (R_g - R)^2) + \phi_{eff}(R_g, 0)$$

$$\dot{R} = \frac{\partial H}{\partial P_R} = \frac{P_R}{m^2} \quad \dot{P}_R = -\frac{\partial H}{\partial R} = -k^2 \frac{(R_g - R)}{x}$$

$$\dot{z} = \frac{\partial H}{\partial P_z} = \frac{P_z}{m^2} \quad \dot{P}_z = -\frac{\partial H}{\partial z} = -v^2 z$$

b) $\ddot{P}_R = m\ddot{R} \quad m\ddot{R} = -k^2(R_g - R)$

$$m\ddot{x} = -k^2 x$$

$$\ddot{x} + \frac{k^2}{m} x = 0$$

$$\ddot{P}_z = m\ddot{z}$$

$$m\ddot{z} = -v^2 z$$

$$\ddot{z} + \frac{v^2}{m} z = 0$$

Ecuações del movimiento
armónico simple en dos
coordenadas
 R y z
con frecuencias de $\frac{k^2}{m}$
y $\frac{v^2}{m}$ re

c) H_R y H_z hamiltonianos de un movimiento armónico de una estrella en su órbita de equilibrio y ϕ_{eff} potencial efectivo gravitacional

2) Si se hace simetría axial

$$\phi_{eff} = \phi + \frac{L_z^2}{2R^2} \quad \text{con } v_c^2(R) = R \frac{\partial \phi}{\partial R}$$

$$k^2 = \frac{\partial^2 \phi}{\partial R^2} + \frac{3L_z^2}{Rg^4}$$

$$v_c^2 = \frac{GM}{R}$$

$$v^2 = \frac{\partial^2 \phi_{eff}}{\partial z^2}$$

$$\frac{L_z^2}{R^2} = v_c^2$$

de la ecuación de poisson

$$4\pi G\rho = \nabla^2 \phi = \frac{1}{R} \frac{\partial}{\partial R} \left(R \frac{\partial \phi_{eff}}{\partial R} \right) + \frac{\partial^2 \phi_{eff}}{\partial z^2}$$

$$4\pi G\rho = \frac{1}{R} \frac{\partial}{\partial R} \left(R \frac{\partial \phi}{\partial R} + \frac{L_z^2}{R^2} \right) + v^2$$

$$4\pi G\rho = \frac{1}{R} \left(\frac{\partial \phi}{\partial R} + R \frac{\partial^2 \phi}{\partial R^2} + v^2 \right)$$

$$4\pi G\rho = \frac{1}{R} \left(\frac{\partial \phi}{\partial R} + R k^2 - \frac{3L_z^2}{R^4} + v^2 \right)$$

$$\rho = \frac{1}{4\pi G} \left(\frac{1}{R} \frac{\partial \phi}{\partial R} + 3 \frac{v_c^2}{R^4} + \frac{v^2}{R} \right)$$